

Module 01: Introduction to Agentic AI & AI Agents

Virtual Environment Setup on VS Code (macOS)

Hands-on: Setting Up a Python Virtual Environment on VS Code (macOS)

Why Use a Virtual Environment?

A virtual environment helps you:

- Prevent dependency conflicts between different Python projects.
 - Keep project packages isolated from the global Python installation.
 - Easily share projects by maintaining a list of required packages.
 - Ensure consistent development across different systems.
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Step-by-Step Guide

Step 1: Open the Terminal

- Open **Terminal** using one of the following methods:
 - Press **Command** (⌘) + **Space**, type **Terminal**, and press **Enter**.
 - Open **Applications** → **Utilities** → **Terminal**.
-

Step 2: Navigate to the Desired Location

Choose the location where you want to create your project.

Example:

```
cd ~/Documents
```

You may replace `Documents` with any folder of your choice.

Step 3: Create a Project Folder

Create a new folder named **test1**.

```
mkdir test1
```

Step 4: Enter the Project Folder

Move into the project directory.

```
cd test1
```

Verify your current directory:

```
pwd
```

Step 5: Launch VS Code

Open the current folder in VS Code.

```
code .
```

Note:

If the `code` command is not available:

1. Open VS Code.
2. Press **Command + Shift + P**.
3. Search for:

```
Shell Command: Install 'code' command in PATH
```

1. Select the command.
 2. Restart Terminal.
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Step 6: Verify the Project Folder

In VS Code:

- Open the **Explorer** panel.

- Ensure the folder **test1** appears as the root project directory.
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Step 7: Create a Python File

Inside VS Code:

- Click **New File**
- Name it:

```
new_env_test.py
```

Step 8: Open the Integrated Terminal

In VS Code:

- Select **Terminal → New Terminal**

or

Press

```
Control + `
```

(macOS uses the default Terminal shell such as **zsh**.)

Step 9: Check Python Installation

Verify Python is installed.

```
python3 --version
```

Example output:

```
Python 3.12.x
```

If Python is not installed, install it first before continuing.

Step 10: Create the Virtual Environment

Run:

```
python3 -m venv env1
```

This creates a folder named:

```
env1
```

which contains the isolated Python environment.

Step 11: Activate the Virtual Environment

Run:

```
source env1/bin/activate
```

The terminal prompt changes from:

```
MacBook:test1 username$
```

to

```
(env1) MacBook:test1 username$
```

This confirms that the virtual environment is active.

Step 12: Select the Python Interpreter

Open the Command Palette.

Press:

```
Command + Shift + P
```

Search for:

Python: Select Interpreter

Choose:

./env1/bin/python

This ensures VS Code uses the virtual environment for running Python code and installing packages.

Step 13: Create a Sample Python Program

Create the following program inside **new_env_test.py**

```
def add(x, y):  
    return x + y  
  
def subtract(x, y):  
    return x - y  
  
def multiply(x, y):  
    return x * y  
  
def divide(x, y):  
    if y == 0:  
        return "Division by zero!"  
    return x / y  
  
print("Addition:", add(5,3))  
print("Subtraction:", subtract(5,3))  
print("Multiplication:", multiply(5,3))  
print("Division:", divide(5,3))
```

Step 14: Run the Python Program

Run:

python new_env_test.py

or

```
python3 new_env_test.py
```

Expected output:

```
Addition: 8  
Subtraction: 2  
Multiplication: 15  
Division: 1.6666666666666667
```

Step 15: Create a Requirements File

Create a new file named:

```
requirements.txt
```

Add the following:

```
pandas  
numpy
```

Step 16: Install Required Packages

Run:

```
pip install -r requirements.txt
```

If needed:

```
pip3 install -r requirements.txt
```

The required packages are installed inside the virtual environment.

Step 17: Verify the Installation

Replace the contents of **new_env_test.py** with:

```
import numpy as np
import pandas as pd

arr = np.array([1,2,3,4,5])

print("NumPy Array:")
print(arr)

data = {
    "Name":["Alice","Bob","Charlie"],
    "Age":[25,30,35]
}

df = pd.DataFrame(data)

print("\nPandas DataFrame:")
print(df)
```

Run:

```
python new_env_test.py
```

Expected Output:

```
NumPy Array:
[1 2 3 4 5]

Pandas DataFrame:
   Name  Age
0  Alice   25
1   Bob   30
2 Charlie   35
```

Deactivating the Virtual Environment

When finished working:

```
deactivate
```

The terminal prompt returns to its normal state.

Summary

In this module, you learned how to:

- Open Terminal on macOS
- Create a project folder
- Open the project in VS Code
- Create a Python virtual environment
- Activate the virtual environment
- Select the correct Python interpreter
- Create and run Python programs
- Install packages using `requirements.txt`
- Verify package installation
- Deactivate the virtual environment

Following these steps ensures each Python project has its own isolated environment, making development cleaner, reproducible, and easier to manage.